History of NLP

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GALLOGLY COLLEGE OF ENGINEERING
SCHOOL OF COMPUTER SCIENCE
The UNIVERSITY of OKLAHOMA

iSAT
NSF AI Institute for Student-AI Teaming

University of Colorado Boulder

THE UNIVERSITY OF UTAH

Amazon AI

Baidu 百度

Natural Language Processing

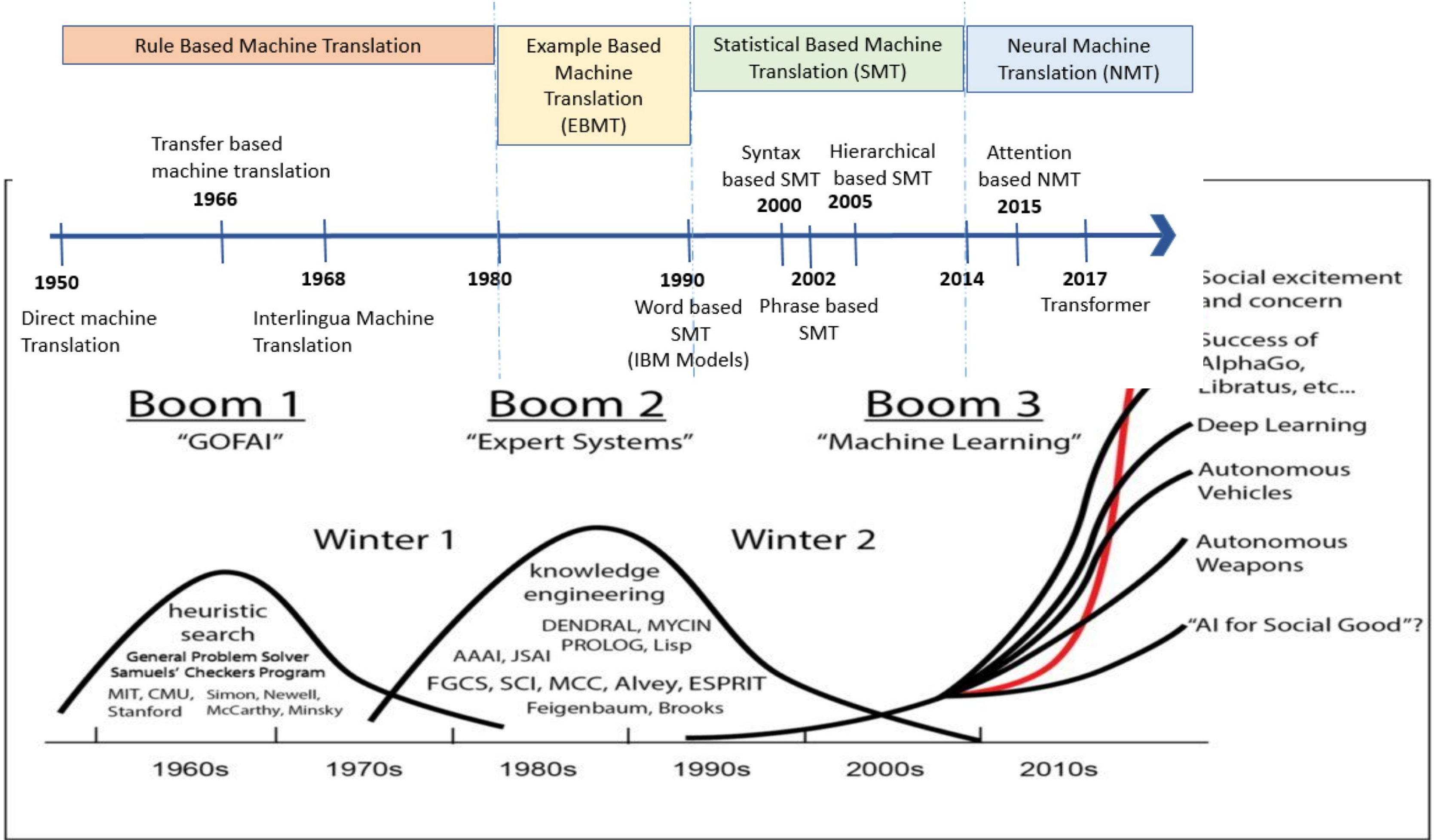
<u>NLP</u> is the study what goes into getting computers to perform <u>*useful*</u> and <u>*interesting*</u> tasks involving human language.

<u>Computational linguistics</u> is the use of computer models to provide <u>*insight*</u> into the nature of human language.

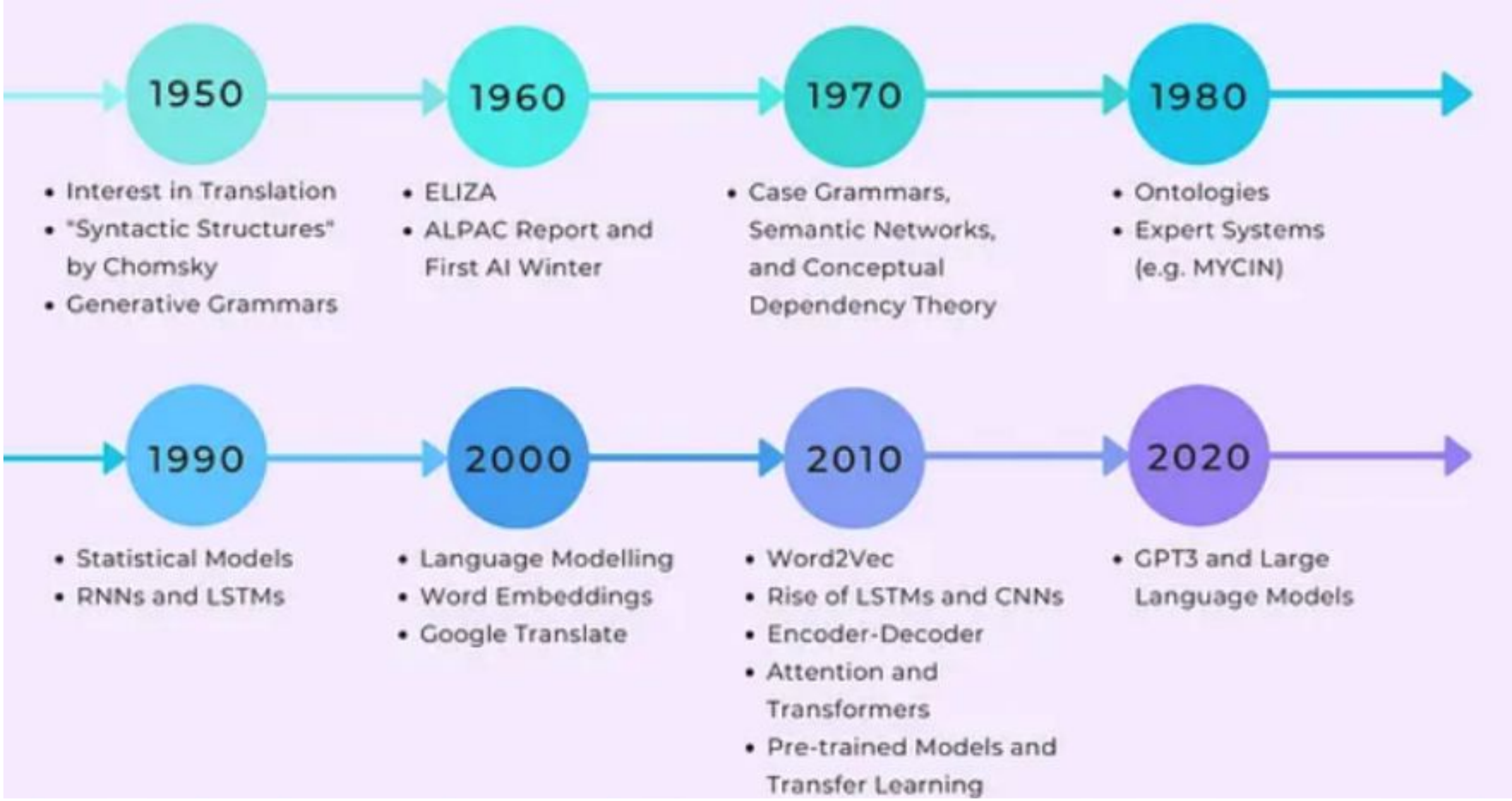
Linguistics(Theory)

https://aclanthology.org/

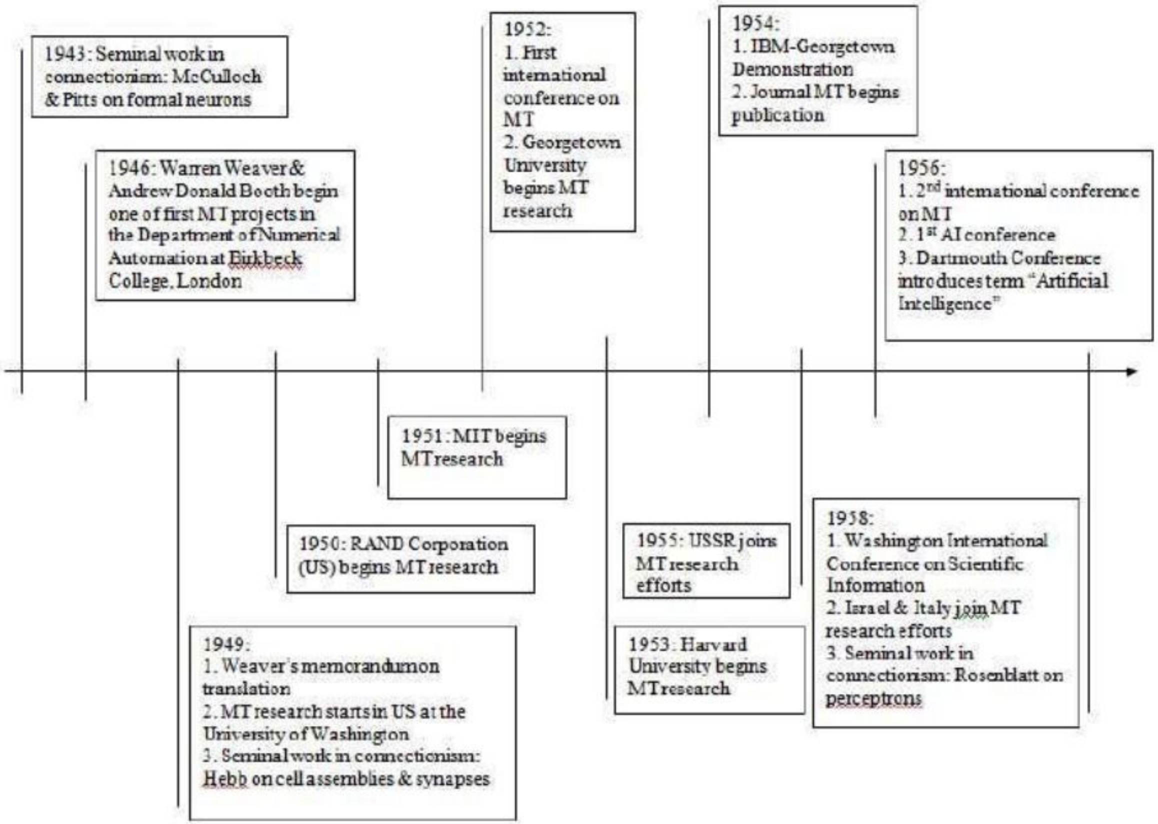
AI Winters(1974–1980, 1987–2000, ?)

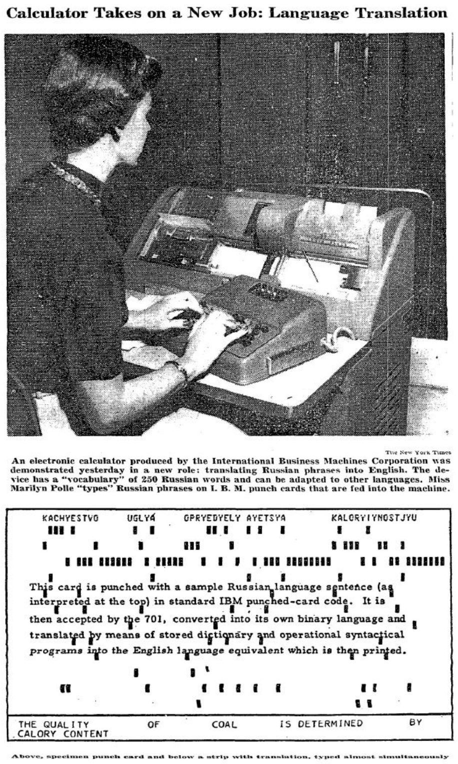


The Brief History of NLP (1974–1980, 1987–2000, ?)



Machine Translation (1950-60s, Rule-based)





Calculator Takes on a New Job: Language Translation

The New York Times

An electronic calculator produced by the International Business Machines Corporation was demonstrated yesterday in a new role: translating Russian phrases into English. The device has a "vocabulary" of 250 Russian words and can be adapted to other languages. Miss Marilyn Polle "types" Russian phrases on I. B. M. punch cards that are fed into the machine.

KACHYESTVO UGLYÁ OPRYEDYELYAYETSYA KALORYIYNOSTJYU

This card is punched with a sample Russian language sentence (as interpreted at the top) in standard IBM punched-card code. It is then accepted by the 701, converted into its own binary language and translated by means of stored dictionary and operational syntactical programs into the English language equivalent which is then printed.

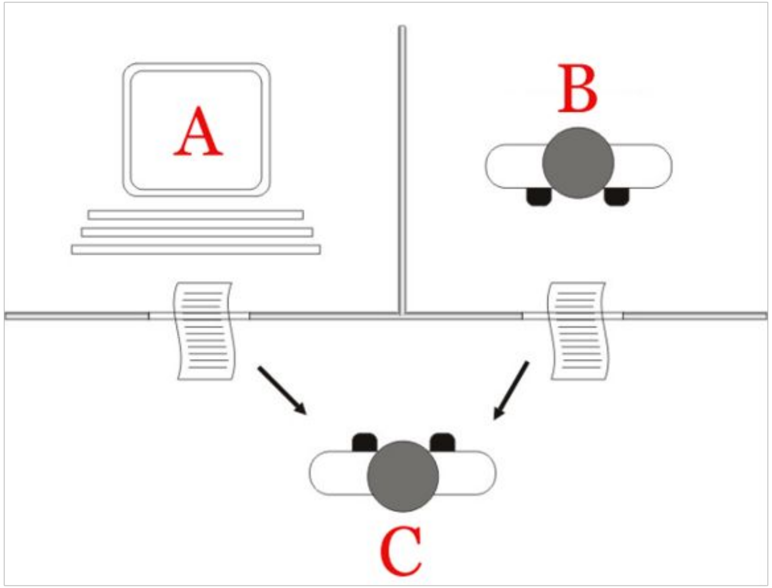
THE QUALITY OF COAL IS DETERMINED BY CALORY CONTENT

Above, specimen punch card and below a strip with translation, typed almost simultaneously

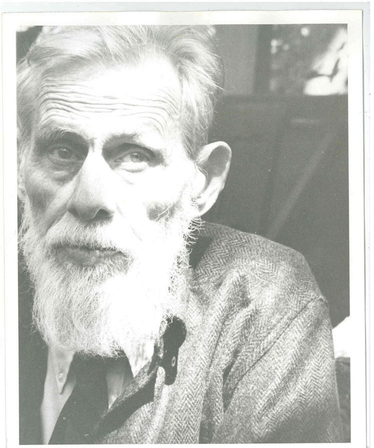
Turing Test: 1950

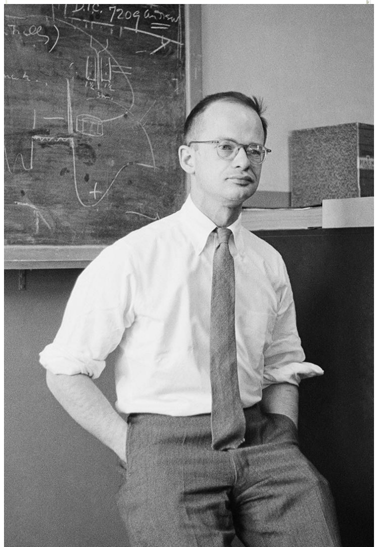
Turing Test

- Imitation game with 3 participants: A) a computer, B) a human player, and C) an interrogator.
- Goal for C is to determine which one is which
- Goal for A is to fool C into believing A is the person



1943: “McCulloch–Pitts” neuron, Threshold Unit

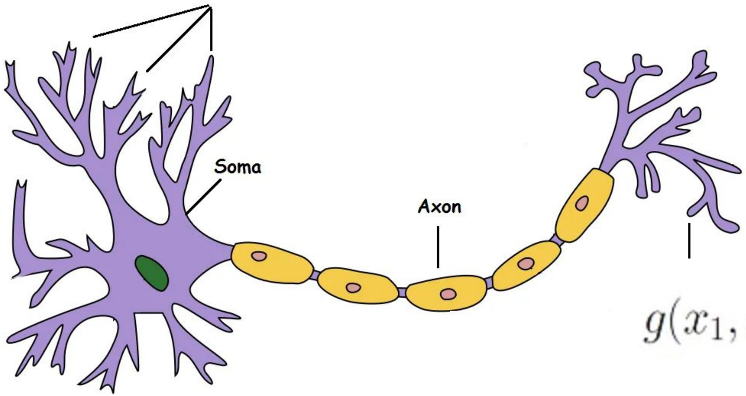


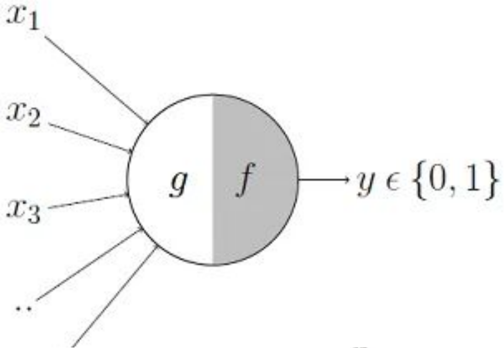


A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY

WARREN S. MCCULLOCH AND WALTER PITTS

FROM THE UNIVERSITY OF ILLINOIS, COLLEGE OF MEDICINE, DEPARTMENT OF PSYCHIATRY AT THE ILLINOIS NEUROPSYCHIATRIC INSTITUTE, AND THE UNIVERSITY OF CHICAGO

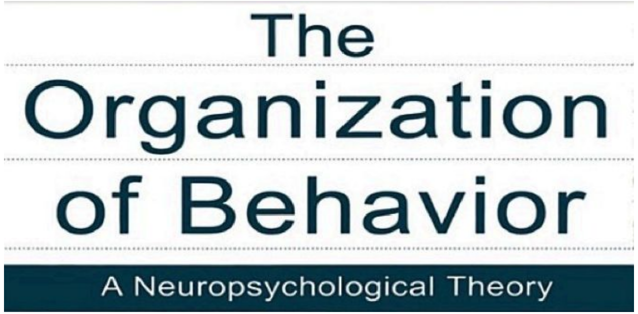




$$g(x_1, x_2, x_3, \ldots, x_n) = g(\mathbf{x}) = \sum_{i=1}^{n} x_i$$

$$\begin{aligned} y = f(g(\mathbf{x})) &= 1 \quad if \quad g(\mathbf{x}) \geq \theta \\ &= 0 \quad if \quad g(\mathbf{x}) < \theta \end{aligned}$$

1949: Donald O. Hebb



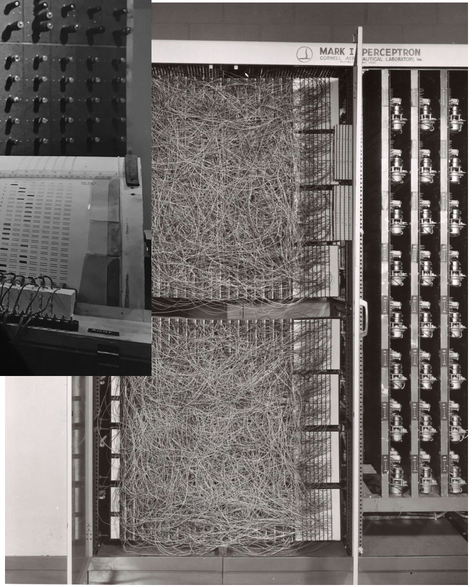
functions. The decade ended with the publication of Donald Hebb's book *The Organization of Behavior*, the first attempt to base a large-scale theory of psychology on conjectures about neural networks. The central idea of Hebb's book was that such networks might learn by constructing internal representations of concepts in the form of what Hebb called "cell-assemblies"—subfamilies of neurons that would learn to support one another's activities. There had been earlier attempts to explain psychology in terms of "connections" or "associations," but (perhaps because those connections were merely between symbols or ideas rather than between mechanisms) those theories seemed *then* too insubstantial to be taken seriously by theorists seeking models for mental mechanisms. Even after Hebb's proposal, it was years before research in artificial intelligence found suitably convincing ways to make symbolic concepts seem concrete.

Symbolism vs. Connectionism

1954
Georgetown experiment
Machine could translate
250 words and
6 **grammar rules**

1957: Mark I Perceptron machine

Cornell **Aeronautical** Laboratory
Frank Rosenblatt



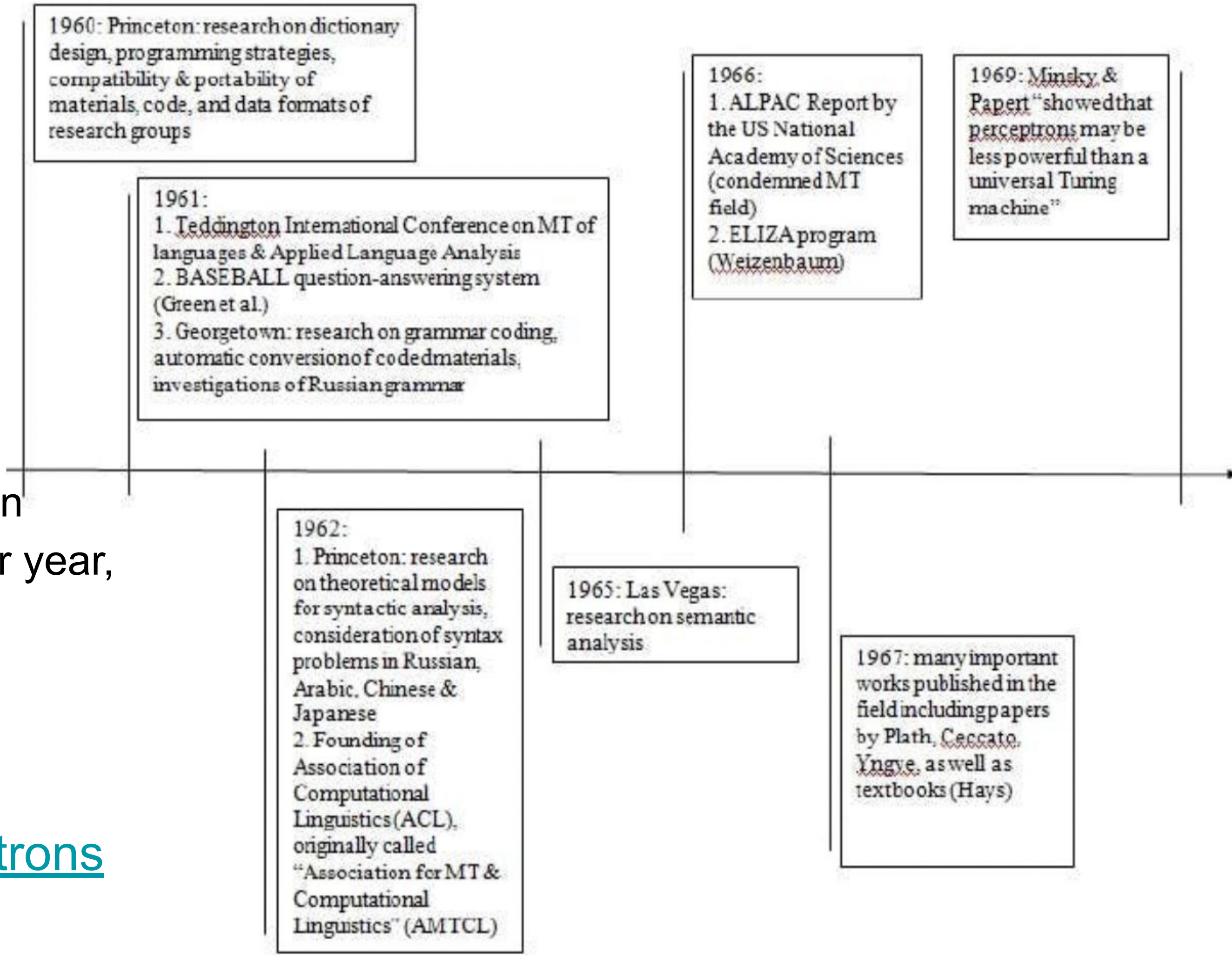
NLP in the 1960s

The begging of the First AI Winter!

1966 ALPAC report:

- only $20 million spent on translation in the US per year, mainly on machines
- no point in machine translation

1969 criticism of [perceptrons]



1970-1992 Hand-built Demonstration NLP Systems.

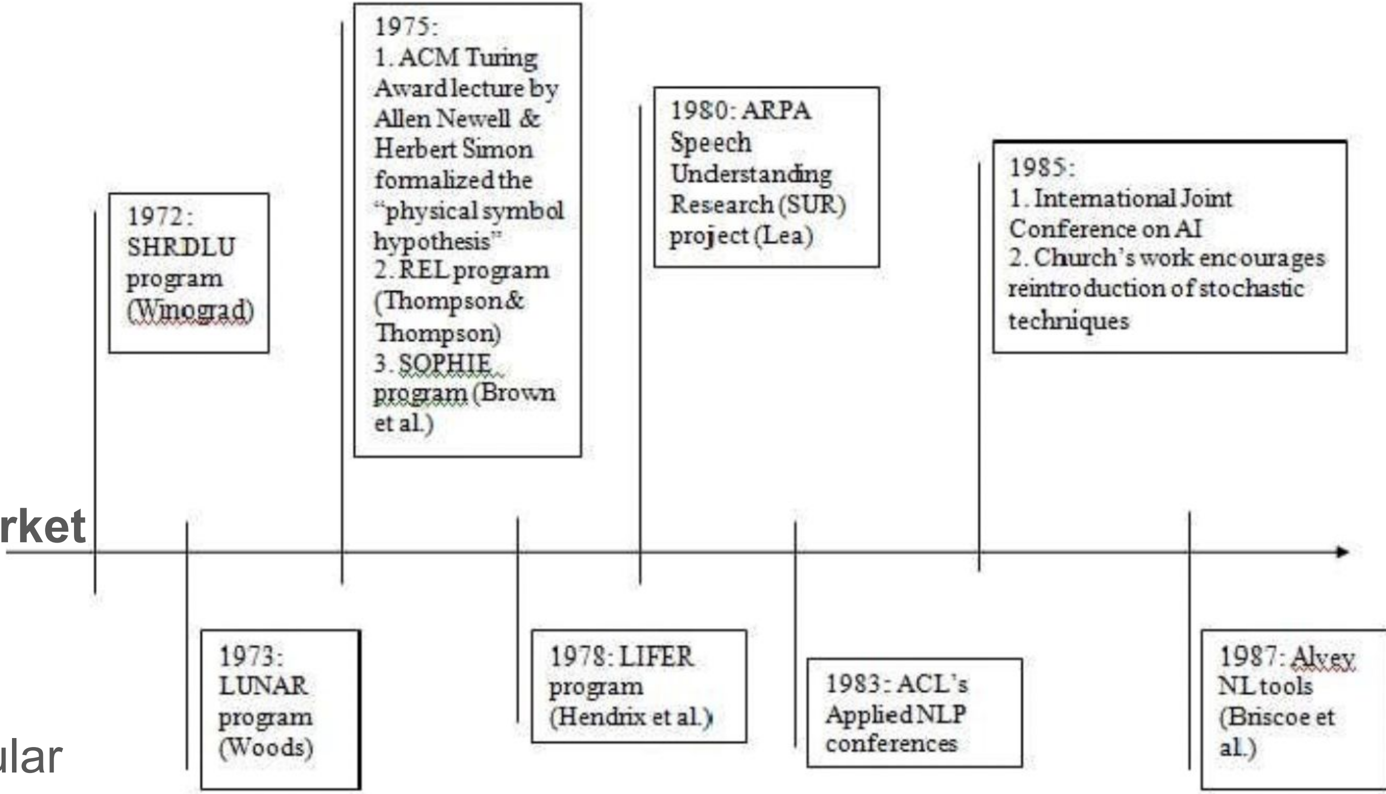
Expert Systems(Lisp Machines):

XCON, for eXpert CONfigurer(1980-86)

- automatically selecting the computer system components
- saved **40 million dollars over just six years of operation**

1987 **Lisp Machine Market Collapse**

Connectionism also developed, but not popular



1993-2012, Statistical or Probabilistic NLP and then more general Supervised ML for NLP

1. Counting on Words: Text Mining, didn't go far for language understanding
2. Linguistic Language Understanding
   a. **Corpus Linguistic**
      i. **WordNet**
      ii. Syntax:Penn treeBank
      iii. Semantics:FrameNet/VerbNet/Prop Bank/AMR/UCCA/SDP
   b. Supervised Learning for NLP Pipeline

2006-2009, ImageNet
2010-2019, ImageNet Challenge

Synsets/Concepts Relations

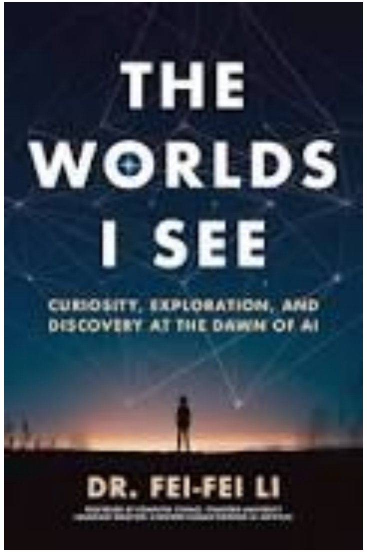
PRINCETON UNIVERSITY

WordNet

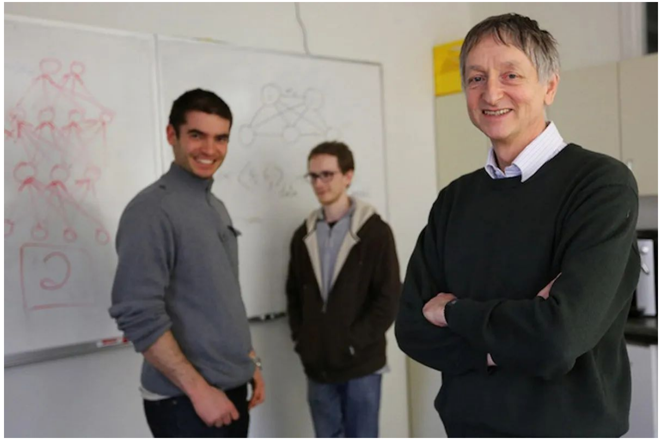
A Lexical Database for English

IMAGENET

https://observablehq.com/@mbostock/imagenet-hierarchy



Sept'2012 AlexNet: Reimplementing/Improving LeNET with GPU



ImageNet Classification with Deep Convolutional Neural Networks

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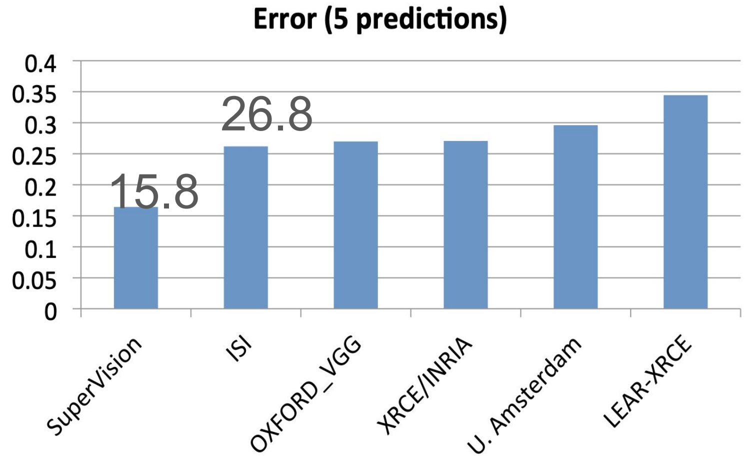

- **Deeper(7 =>9) and bigger(60k=>60m) LeNet(1989-1998)**
- Key modifications
  - Dropout (hinton'2012, Axriv=>2014 JMLR)
  - Tanh => ReLu (training)
  - AvgPooling =>MaxPooling

Deep Learning Worked!!!

2008 Or Even Earlier, Neural Networks using GPU Computing

Andrew NG

Latent Dirichlet Allocation (ICML'03)

coursera

DeepLearning.AI

Google Brain

AIFUND

LandingAI

Large-scale Deep Unsupervised Learning using Graphics Processors

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Computer Science Department, Stanford University, Stanford CA 94305 USA

Debate

Model-centric AI:

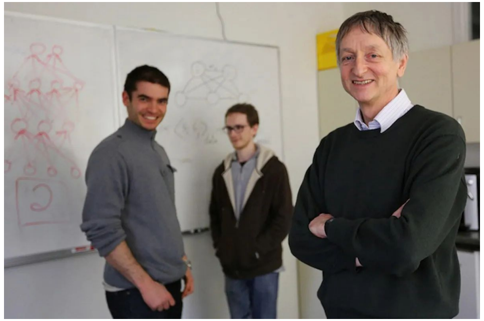
- Given a dataset, try to produce the best model (think 6.036/6.390)
- Change the model to improve performance on an AI task
  - e.g. modify the loss function, hyper-parameters, etc.

Data-centric AI(Andrew NG):

- Given any model, try to improve the training dataset
- **Systematically/algorithmically** change the dataset to improve performance on an AI task

Dec 2012, Secret Auction at NIPS(NeurIPS'2012)

DNNResearch



Baidu, 12 million => 40 million the end of the first day

DeepMind (Dropout first)

Microsoft (Dropout Second)

Ilya

- Word2Vec
- TensorFlow
- Seq2Seq
- AlphaGo

Dec 2015-May 2024 OpenAI

- GPT-Family
- SuperAlignment…

June 2024- ?

Safe Superintelligence Inc.

Update: We've raised $1B from NFDG, a16z, Sequoia, DST Global, and SV Angel. See more here.

Google Brain

44 millions

Alex

2017

Dessa

Geoffrey

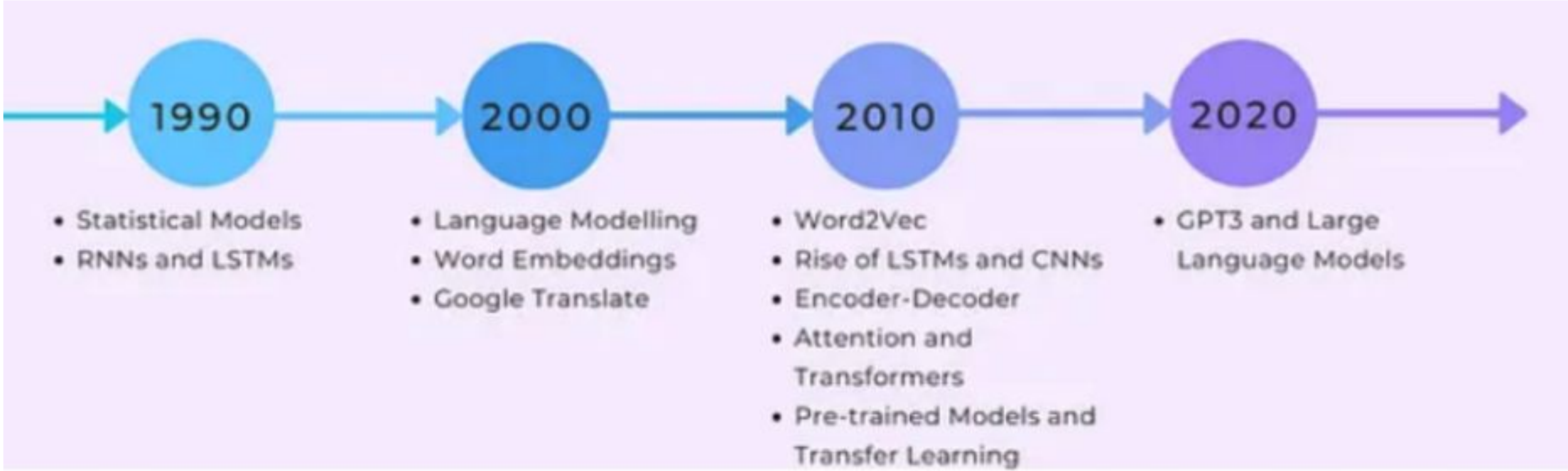
- Fast-weight
- Capsule network
- Forward-Forward
- Analog bits
- Neural Additive Models
- Pix2Seq

"freely speak out about the risks of A.I.

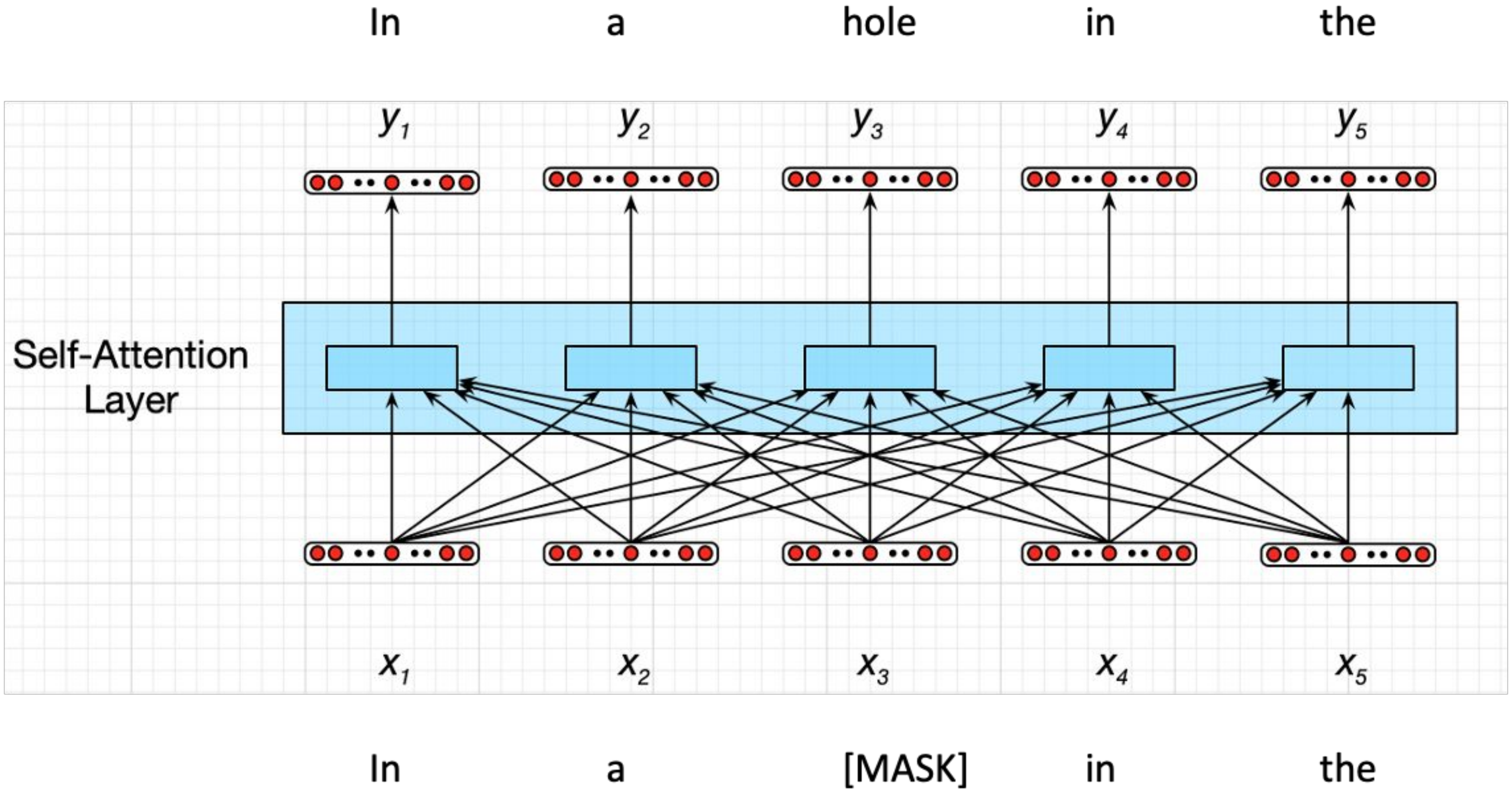
Left Google at 2023

- Word Embedding
- Attention Mechanism
- Attention is all you need
- Pretrained Transformers
- **Large** Language Models

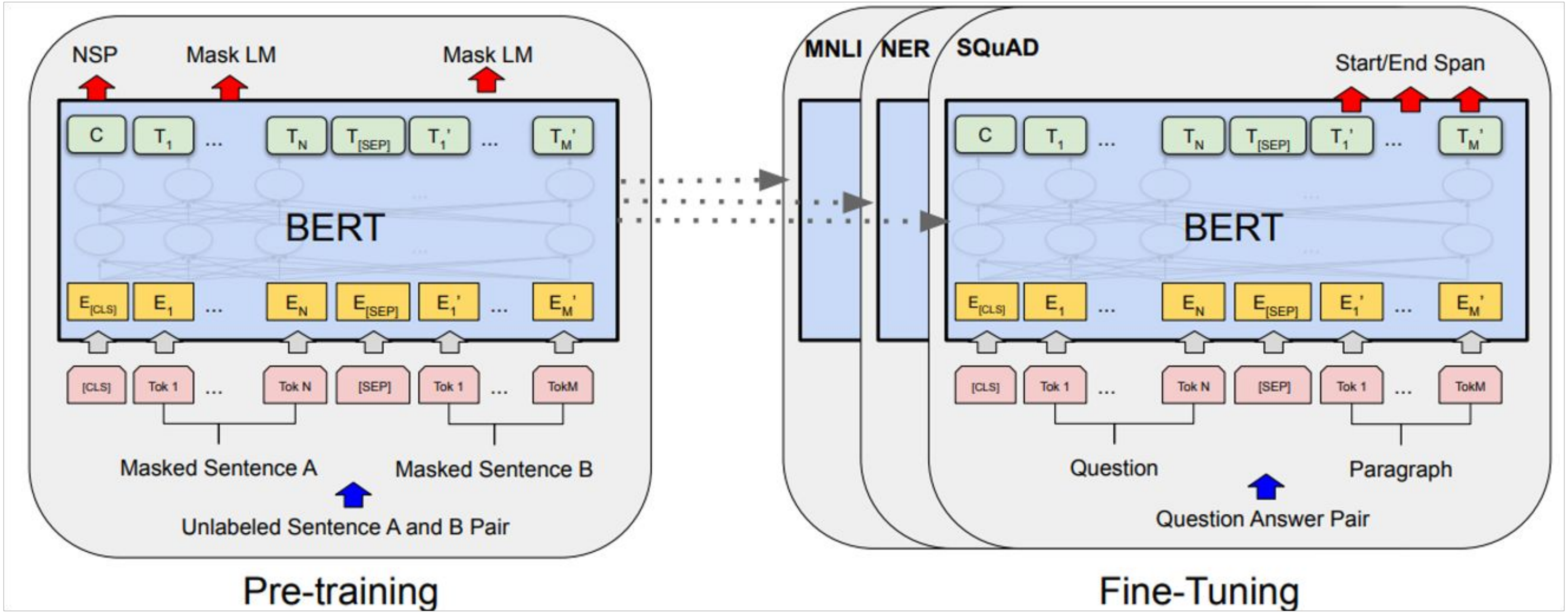
Deep Learning Worked, got **predictably better with scale**, and we dedicated increasing resources to it.



Transformer(Personally, I think it is also Neuro-Symbolic)



Pretraining-Finetuning

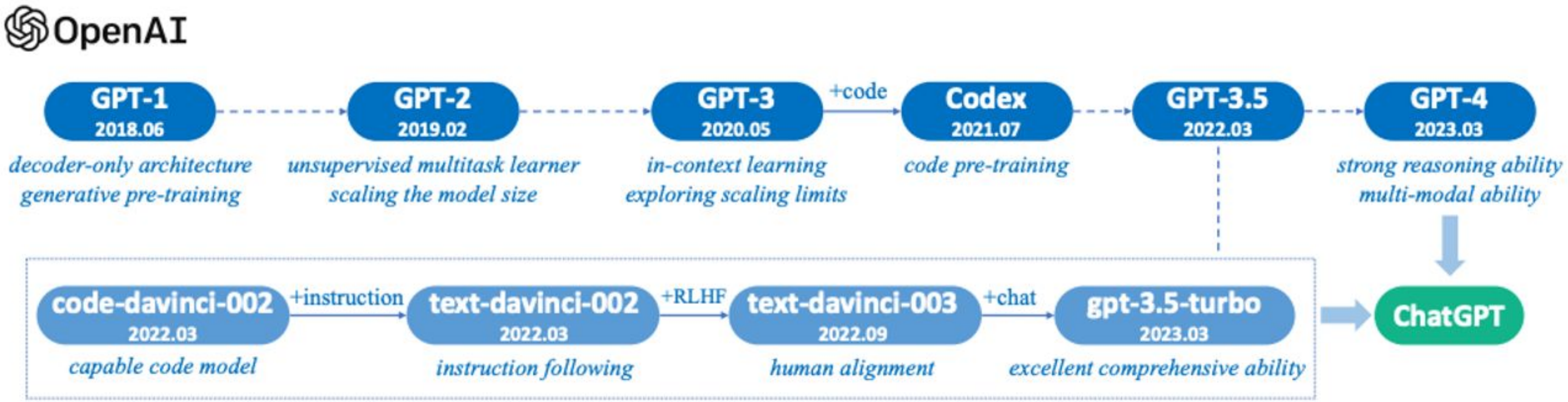


GPT-2 => GPT 3: Scaling Law of Self-Supervised

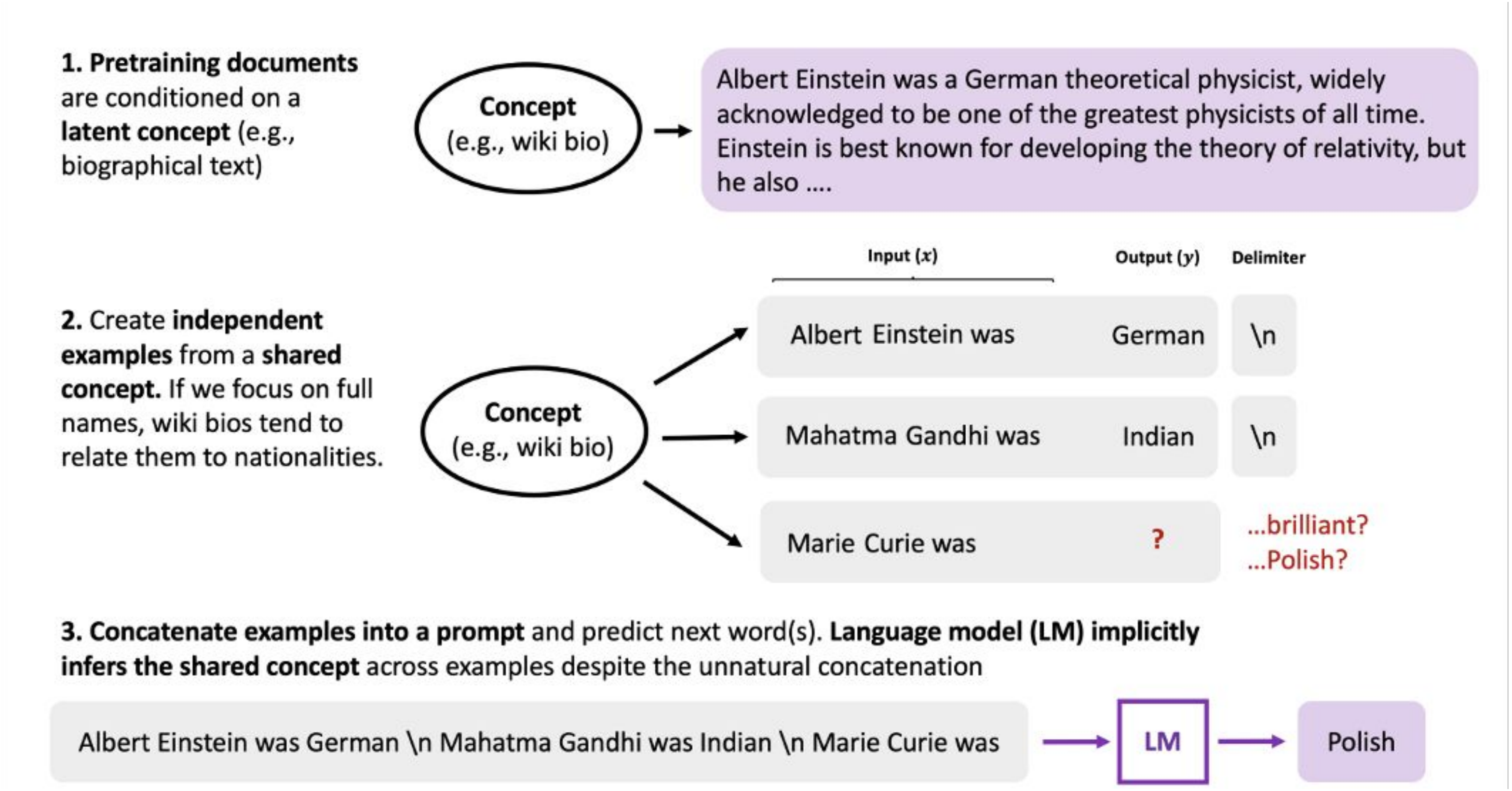
Open-AI published a long technique report

It helps other companies to build GPT-3 like models

Scaling Law: DeepMind first, OpenAI All In Scaling Law



In-Context Learning (Connecting with Learned Concepts)



Latent prompt(task) concept: **Nationality**

$$p(\text{output}|\text{prompt}) = \int_{\text{concept}} p(\text{output}|\text{concept}, \text{prompt})p(\text{concept}|\text{prompt})d(\text{concept})$$

Chat-GPT Moment: Instruction&Alignment

GPT3+RHLF=> RL Transfer

Post-Training Paradigm: Deep Reinforcement

But, you need to write **long prompt to guide the ChatGPT to generate.**

PPO, requiring more computing, not efficient, but could find the good model
DQN, offline RL, efficient with limited resource

Pre-training may be bottlenecked by data and architecture?

We only have enough data

Let’s get to ceiling first!

- Data Cleaning
- MoE(Mistral, GPT-Turbo, Grok-2)
- Multimodality Learning is easier than text-only(GPT4o)

Post-training is still very valuable

Q* Project How many rs are in the word **Strawberry**?

OpenAI o1-preview(Strawberry): Reasoning

- Long Context will be the first step.
- RL may help on **causal inference**, and **reduce hallucination**.

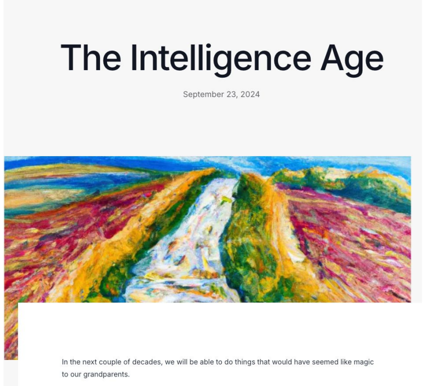
Prompt(Task): How to design tasks;Short Prompt without much guidance
Reward(Feedback): Universal/Generalizable Reward Model?
Exploration: Latency/ Efficient Combinary Search/Latency
What is the new Scaling Law?

AGI?ASI?

OpenAI

- Level 1: Chatbots: what we currently have.
- Level 2: Reasoners: PhD-level problem solvers.
- Level 3: Agents: AI systems that can spend days taking actions for you.
- Level 4: Innovators: your AI version of Thomas Edison.
- Level 5: Organizations: a single AI doing the job of a whole company.

Performance (rows) x **Generality** (columns)	**Narrow** *clearly scoped task or set of tasks*	**General** *wide range of non-physical tasks, including metacognitive tasks like learning new skills*
Level 0: No AI	**Narrow Non-AI** calculator software; compiler	**General Non-AI** human-in-the-loop computing, e.g., Amazon Mechanical Turk
Level 1: Emerging *equal to or somewhat better than an unskilled human*	**Emerging Narrow AI** GOFAI (Boden, 2014); simple rule-based systems, e.g., SHRDLU (Winograd, 1971)	**Emerging AGI** ChatGPT (OpenAI, 2023), Bard (Anil et al., 2023), Llama 2 (Touvron et al., 2023), Gemini (Pichai & Hassabis, 2023)
Level 2: Competent *at least 50th percentile of skilled adults*	**Competent Narrow AI** toxicity detectors such as Jigsaw (Das et al., 2022); Smart Speakers such as Siri (Apple), Alexa (Amazon), or Google Assistant (Google); VQA systems such as PaLI (Chen et al., 2023); Watson (IBM); SOTA LLMs for a subset of tasks (e.g., short essay writing, simple coding)	**Competent AGI** not yet achieved
Level 3: Expert *at least 90th percentile of skilled adults*	**Expert Narrow AI** spelling & grammar checkers such as Grammarly (Grammarly, 2023); generative image models such as Imagen (Saharia et al., 2022) or Dall-E 2 (Ramesh et al., 2022)	**Expert AGI** not yet achieved
Level 4: Virtuoso *at least 99th percentile of skilled adults*	**Virtuoso Narrow AI** Deep Blue (Campbell et al., 2002), AlphaGo (Silver et al., 2016; 2017)	**Virtuoso AGI** not yet achieved
Level 5: Superhuman *outperforms 100% of humans*	**Superhuman Narrow AI** AlphaFold (Jumper et al., 2021; Varadi et al., 2021), AlphaZero (Silver et al., 2018), StockFish (Stockfish, 2023)	**Artificial Superintelligence (ASI)** not yet achieved



https://ia.samaltman.com/

A few thousands days to ASI?

What’s in the future? Famous Debates

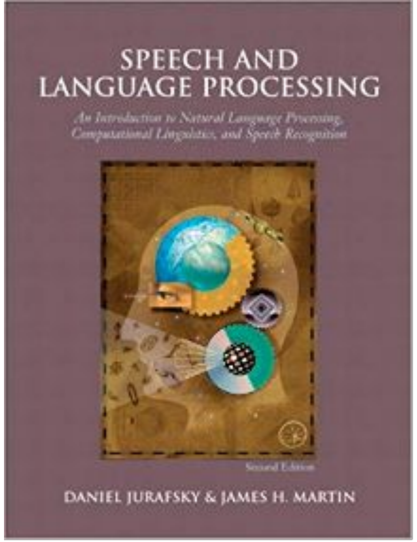
- Connectionism v.s. Symbolicism?
- Deep Learning Overfits, which would not generalize?
- Language Models don’t understand the meaning?
- AGI? ASI?
- One model for all?(Multi-task learning indeed helps)
- Data-Centric ML and Model-Centric ML?
- Hope or Horror?

Data, Model, Computing

Data-centric ML
Neuro-Symbolic?
Analog Computers?Quantum-Computing?
Forward-Forward?
Fast-Weights?
Multimodal Learning?

Advertisements

Natural Language Processing (Sp’2025)
(Text Analytics, CS’5293)



Hiring! I'm seeking enthusiastic students who are passionate about exploring NLP, with a focus on

- **Dialogue**
- **Structured Prediction**
- **Complex Reasoning**
- **Multimodal Learning**